

1. A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2}m\omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of n^{th} orbit will be proportional to

[2023 (06 Apr Shift 2)]

- (1) $\sqrt{n}$
(2) $\frac{1}{n}$
(3) n^2
(4) n

2. A body is dropped on ground from a height h_1 and after hitting the ground, it rebounds to a height h_2 . If the ratio of velocities of the body just before and after hitting ground is 4, then percentage loss in kinetic energy of the body is $\frac{x}{4}$. The value of x is _____.

[2023 (06 Apr Shift 2)]

3. The momentum of a body is increased by 50%. The percentage increase in the kinetic energy of the body is _____%.

[2023 (08 Apr Shift 1)]

4. A bullet of mass 0.1 kg moving horizontally with speed 400 m s⁻¹ hits a wooden block of mass 3.9 kg kept on a horizontal rough surface. The bullet gets embedded into the block and moves 20 m before coming to rest. The coefficient of friction between the block and the surface is _____.

[2023 (08 Apr Shift 2)]

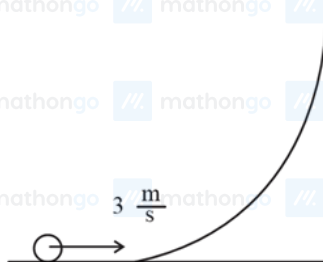
- (1) 0.90
(2) 0.50
(3) 0.65
(4) 0.25

5. A body of mass 5 kg is moving with a momentum of 10 kg m s⁻¹. Now a force of 2 N acts on the body in the direction of its motion for 5 s. The increase in the Kinetic energy of the body is _____ J.

[2023 (08 Apr Shift 2)]

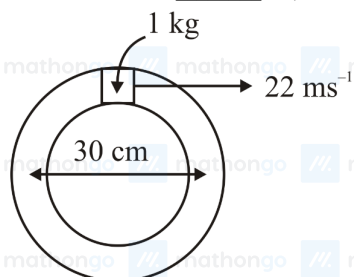
6. A hollow spherical ball of uniform density rolls up a curved surface with an initial velocity 3 m s⁻¹ (as shown in figure). Maximum height with respect to the initial position covered by it will be _____ cm

(take, $g = 10 \text{ m s}^{-2}$)



[2023 (08 Apr Shift 2)]

7. A closed circular tube of average radius 15 cm, whose inner walls are rough, is kept in vertical plane. A block of mass 1 kg just fit inside the tube. The speed of block is 22 m s⁻¹, when it is introduced at the top of tube. After completing five oscillations, the block stops at the bottom region of tube. The work done by the tube on the block is _____ J. (Given $g = 10 \text{ m s}^{-2}$).



[2023 (10 Apr Shift 1)]

8. If the maximum load carried by an elevator is 1400 kg (600 kg-Passengers + 800 kg-elevator), which is moving up with a uniform speed of 3 m s⁻¹ and the frictional force acting on it is 2000 N, then the maximum power used by the motor is _____ kW. ($g = 10 \text{ m s}^{-2}$)

[2023 (10 Apr Shift 2)]

9. A force $\vec{F} = (2 + 3x)\hat{i}$ acts on a particle in the x direction where F is in Newton and x is in meter. The work done by this force during a displacement from $x = 0$ to $x = 4 \text{ m}$ is _____ J.

[2023 (11 Apr Shift 1)]

10. A block of mass 5 kg starting from rest pulled up on a smooth incline plane making an angle of 30° with horizontal with an effective acceleration of 1 m s^{-2} . The power delivered by the pulling force at $t = 10 \text{ s}$ from the start is _____ W.
[Use $g = 10 \text{ m s}^{-2}$]
(Calculate the nearest integer value)
[2023 (11 Apr Shift 2)]
11. To maintain a speed of 80 km h^{-1} by a bus of mass 500 kg on a plane rough road for 4 km distance, the work done by the engine of the bus will be _____ kJ.
[The coefficient of friction between tyre of bus and road is 0.04]
[2023 (12 Apr Shift 1)]
12. The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x+1}}$, that are capable of raising 300 kg water in 5 minutes and 50 kg water in 2 minutes respectively from a well of 100 m deep. The value of x will be
[2023 (13 Apr Shift 1)]
(1) 16
(2) 2
(3) 2.4
(4) 4
13. A body of mass $(5 \pm 0.5) \text{ kg}$ is moving with a velocity of $(20 \pm 0.4) \text{ m s}^{-1}$. Its kinetic energy will be
[2023 (13 Apr Shift 1)]
(1) $(1000 \pm 0.14) \text{ J}$
(2) $(500 \pm 0.14) \text{ J}$
(3) $(500 \pm 140) \text{ J}$
(4) $(1000 \pm 140) \text{ J}$
14. A car accelerates from rest of $u \text{ m s}^{-1}$. The energy spent in this process is $E \text{ J}$. The energy required to accelerate the car from $u \text{ m s}^{-1}$ to $2u \text{ m s}^{-1}$ is $nE \text{ J}$. The value of n is _____.
[2023 (13 Apr Shift 2)]
15. A body is released from a height equal to the radius (R) of the earth. The velocity of the body when it strikes the surface of the earth will be: (Given g = acceleration due to gravity on the earth.)
[2023 (15 Apr Shift 1)]
(1) $\sqrt{2gR}$
(2) $\sqrt{gR}$
(3) $\sqrt{4gR}$
(4) $\sqrt{\frac{gR}{2}}$
16. A block of mass 10 kg is moving along x -axis under the action of force $F = 5x \text{ N}$. The work done by the force in moving the block from $x = 2 \text{ m}$ to 4 m will be _____ J.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (1) 2. (375) 3. (125) 4. (4) 5. (30) 6. (75) 7. (245) 8. (48)
9. (32) 10. (300) 11. (784) 12. (1) 13. (4) 14. (3) 15. (2) 16. (30)

1. (1)

The data given is

$$U = \frac{1}{2} m \omega^2 r^2$$

From Bohr's quantization,

$$mvr = \frac{nh}{2\pi}$$

$$\Rightarrow v^2 = \left(\frac{nh}{2\pi m r} \right)^2 \dots (i)$$

In an orbit the value of the kinetic energy is half of the potential energy,

$$K = \frac{m\omega^2 r^2}{4}$$

Substituting the value of equation (i) in the kinetic energy,

$$K = \frac{1}{2} m \left(\frac{n^2 h^2}{4\pi^2 m^2 r^2} \right) = \frac{m\omega^2 r^2}{4}$$

$$\Rightarrow \frac{n^2 h^2}{2\pi^2 m^2 \omega^2} = r^2$$

$$\Rightarrow r \propto \sqrt{n}$$

2. (375)

The given ratio of the velocities is

$$\frac{v_1}{v_2} = 4$$

The percentage loss in kinetic energy is

$$\frac{\frac{mv_1^2}{2} - \frac{mv_2^2}{2}}{\frac{mv_1^2}{2}} \times 100 = \frac{x}{4}$$

$$\Rightarrow \frac{v_1^2 - v_2^2}{v_1^2} = \frac{x}{400}$$

$$\Rightarrow 1 - \frac{v_2^2}{v_1^2} = \frac{x}{400}$$

$$\Rightarrow 1 - \frac{1}{16} = \frac{x}{400}$$

$$\Rightarrow x = 375$$

3. (125)

The relation of kinetic energy and momentum is given by

$$K = \frac{p^2}{2m}$$

The new value of momentum is $p' = 1.5p$

The new value of kinetic energy is

$$\Rightarrow K' = \frac{(1.5p)^2}{2m} = 2.25K$$

Hence, the percentage increase in kinetic energy is

$$\frac{K' - K}{K} \times 100 = \frac{2.25K - K}{K} = 125\%$$

4. (4)

Let the mass of the bullet be m and mass of the block be M . It is given that $V = 400 \text{ m s}^{-1}$. By conservation of momentum,

$$mV = (m + M)v$$

$$\Rightarrow 0.1 \times 400 = (0.1 + 3.9)v$$

$$\Rightarrow v = 10 \text{ m s}^{-1}$$

The force acting on the block

$$f = \mu m'g$$

$$\Rightarrow a = \mu g$$

$$\text{Using } v^2 - u^2 = -2aS$$

$$\Rightarrow 0^2 - 10^2 = 2(-\mu g)(S)$$

$$\Rightarrow \mu = \frac{100}{2 \times 10 \times 20} = 0.25$$

5. (30)

The formula to calculate the change on momentum of the object is given by

$$p_f - p_i = \int F dt \dots (1)$$

Substitute the values of the known parameters into equation (1) and solve to calculate the final momentum of the object.

$$p_f - 10 \text{ kg m s}^{-1} = 2 \text{ N} \times 5 \text{ s}$$

$$\Rightarrow p_f = (10 + 10) \text{ kg m s}^{-1} \\ = 20 \text{ kg m s}^{-1}$$

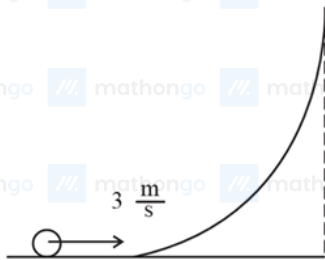
The formula to calculate increase in kinetic energy (ΔK) of the particle is given by

$$\Delta K = \frac{1}{2m} (p_f^2 - p_i^2) \dots (2)$$

Substitute the values of the known parameters into equation (2) to calculate the required increase in kinetic energy.

$$\Delta K = \frac{1}{2 \times 5 \text{ kg}} \times (20^2 - 10^2) \text{ kg}^2 \text{ m}^2 \text{ s}^{-2} \\ = 30 \text{ J}$$

6. (75)



The total kinetic energy of the ball at the bottom of the curve can be calculated as follows:

$$K = \frac{1}{2} M v^2 + \frac{1}{2} \cdot \frac{2}{3} M R^2 \omega^2$$

$$= \frac{1}{2} M v^2 + \frac{1}{3} M (R\omega)^2$$

$$= \frac{1}{2} M v^2 + \frac{1}{3} M v^2 \quad [\because v = \omega R]$$

$$= \frac{5}{6} M v^2 \dots (1)$$

The formula to calculate the potential energy acquired by the ball when it attains the maximum height is given by

$$U = Mgh \dots (2)$$

Equate equation (1) and (2) and simplify to obtain the required height.

$$Mgh = \frac{5}{6} M v^2$$

$$\Rightarrow h = \frac{5v^2}{6g} \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required height attained by the ball.

$$h = \frac{5 \times (3 \text{ m s}^{-1})^2}{6 \times 10 \text{ m s}^{-2}}$$

$$= 0.75 \text{ m} \times \frac{100 \text{ cm}}{1 \text{ m}}$$

$$= 75 \text{ cm}$$

7. (245)

The data given is

$$r = 15 \text{ cm}$$

$$m = 1 \text{ kg}$$

$$v = 22 \text{ m s}^{-1}$$

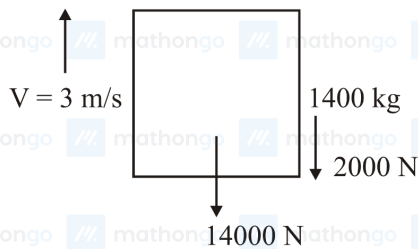
Total work done by the tube on the block is equal to loss in mechanical energy,

$$w = \frac{1}{2} m v^2 + mgh$$

$$\Rightarrow w = \frac{1}{2} \times 1 \times (22)^2 + 1 \times 10 \times 0.3$$

$$\Rightarrow w = 245 \text{ J}$$

8. (48)



The maximum force acting is, $F_{max} = (14000 + 2000) \text{ N}$
 $= 16000 \text{ N}$

The formula for power is given by, $P = F_{max}v$.

It is given, $v = 3 \text{ m s}^{-1}$.

$$\Rightarrow P = 16000 \text{ N} \times 3 \text{ m s}^{-1} = 48000 \text{ W} = 48 \text{ kW}$$

$$\Rightarrow W = 48 \text{ kW}$$

9. (32)

The formula to calculate the work done by a variable force is given by

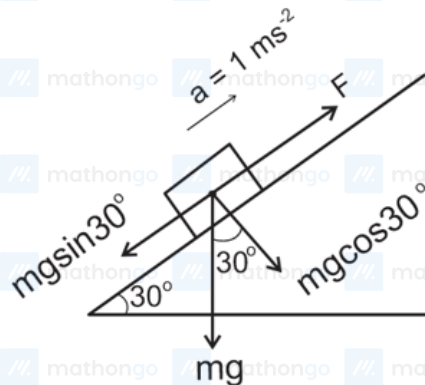
$$W = \int_{x_1}^{x_2} F dx \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required work done.

$$\begin{aligned} W &= \int_0^4 (2 + 3x) dx \\ &= \left[2x + \frac{3}{2}x^2 \right]_0^4 \\ &= [8 + 24] \text{ J} \\ &= 32 \text{ J} \end{aligned}$$

10. (300)

Let's consider the following free-body diagram:



From the above diagram, it can be written that

$$F - mg \sin 30^\circ = ma \dots (1)$$

Substitute the values of the known parameters into equation (1) and solve to calculate the force applied on the block.

$$\begin{aligned} F - 50 \sin 30^\circ &= 5 \times 1 \\ \Rightarrow F &= (5 + 25) \text{ N} \\ &= 30 \text{ N} \end{aligned}$$

The velocity of the block (v) at $t = 10 \text{ s}$ can be written as

$$\begin{aligned} v &= 0 + 1 \text{ m s}^{-2} \times 10 \text{ s} \\ &= 10 \text{ m s}^{-1} \end{aligned}$$

Hence, the required power is given by

$$\begin{aligned} P &= Fv \\ &= 30 \text{ N} \times 10 \text{ m s}^{-1} \\ &= 300 \text{ W} \end{aligned}$$

11. (784)

It is given,

$$v = 80 \text{ km h}^{-1}$$

$$m = 500 \text{ kg}$$

$$S = 4 \text{ km}$$

$$\mu = 0.04$$

The power due to friction is

$$P_f = Fv \text{ where } F = \mu mg$$

$$= \left((0.04)(500 \times 9.8) \times 80 \times \frac{5}{18} \right) \text{ W}$$

Hence, the work done by the engine in overcoming the friction is

$$W = P_f t = P_f \frac{d}{v}$$

$$\Rightarrow W = 20 \times 9.8 \times 80 \times \frac{5}{18} \times \frac{4 \times 10^3}{80 \times \frac{5}{18}} = 784 \text{ kJ}$$

12. (1)

The work done in raising the water is $W = mgh$.

$$\text{Power is } P = \frac{W}{t}$$

Let the powers of both the motors be P_1 & P_2 respectively.

Hence, the ratio of the powers is,

$$\frac{P_1}{P_2} = \frac{\frac{300 \times g \times 100}{5 \times 60}}{\frac{50 \times g \times 100}{2 \times 60}} = \frac{60}{25}$$

$$\Rightarrow \frac{3\sqrt{x}}{\sqrt{x}+1} = 2.4$$

$$\Rightarrow 3\sqrt{x} = 2.4\sqrt{x} + 2.4$$

$$\Rightarrow x = 16$$

13. (4)

The kinetic energy is given by,

$$K = \frac{1}{2}mv^2$$

Differentiating the above equation,

$$dK = \frac{1}{2}(v^2 dm + 2mv dv)$$

Dividing by K ,

$$\Rightarrow \frac{dK}{K} = \frac{dm}{m} + \frac{2dv}{v}$$

$$\Rightarrow \frac{dK}{1000} = \frac{0.5}{5} + \frac{2 \times 0.4}{20}$$

$$\Rightarrow dK = 100 + 40 = 140$$

$$\Rightarrow KE = (1000 \pm 140) \text{ J}$$

14. (3)

The kinetic energy of a moving particle is given by

$$E = \frac{1}{2}mu^2 \dots (1)$$

When the final velocity of the particle becomes twice the initial velocity, the kinetic energy of the particle can be written as

$$\begin{aligned} nE &= \frac{1}{2}m\{(2u)^2 - u^2\} \\ &= \frac{1}{2}m(3u^2) \dots (2) \end{aligned}$$

Divide equation (2) by equation (1) and solve to calculate the value of n .

$$\frac{nE}{E} = \frac{\frac{1}{2}m(3u^2)}{\frac{1}{2}mu^2}$$

$$\Rightarrow n = 3$$

15. (2)

Using conservation of mechanical energy,

$$\begin{aligned} U_i + K_i &= U_f + K_f \\ \Rightarrow -\frac{GMm}{2R} + 0 &= -\frac{GMm}{R} + \frac{1}{2}mv^2 \\ \Rightarrow \frac{1}{2}mv^2 &= \frac{GMm}{2R} \\ \Rightarrow v &= \sqrt{\frac{GM}{R}} = \sqrt{gR} \end{aligned}$$

16. (30)

Work done is $W = \int F dx$

It is given that

$$F = 5x \text{ N}$$

So, work done is

$$\begin{aligned} W &= \int_2^4 5x dx = \left. \frac{5x^2}{2} \right|_2^4 \\ \Rightarrow W &= \frac{5(16-4)}{2} = 30 \text{ J} \end{aligned}$$